

Section 6.1

Inner Product, Length,
and Orthogonality

The (Usual) Inner Product

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, then the (usual) inner product of $\mathbf{u}$ and $\mathbf{v}$ is

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$$

which is a 1×1 matrix, generally regarded as a scalar.

The (usual) inner product of $\mathbf{u}$ and $\mathbf{v}$ is often referred to as the dot product of $\mathbf{u}$ and $\mathbf{v}$.

Example 1

Compute $\mathbf{u} \cdot \mathbf{v}$ and $\mathbf{v} \cdot \mathbf{u}$ for

$$\mathbf{u} = \begin{bmatrix} 1 \\ -3 \\ 7 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 2 \\ 5 \\ -4 \end{bmatrix}.$$

Properties of the Inner Product

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in $\mathbb{R}^n$, and let c be a scalar. Then,

- a. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
- b. $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$
- c. $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$
- d. $\mathbf{u} \cdot \mathbf{u} \geq 0$, and $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$

The Norm of a Vector

The norm of a vector $\mathbf{v}$ in $\mathbb{R}^n$ is the nonnegative scalar

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$$

This gives us the magnitude (length) of $\mathbf{v}$.

Unit Vectors

A vector with norm (length) 1 is a unit vector.

Dividing a vector $\mathbf{v}$ by its norm creates a unit vector in the same direction as $\mathbf{v}$; this is called normalizing the vector.

Example 2

Calculate $\|\mathbf{v}\|$ given $\mathbf{v} = \begin{bmatrix} 6 \\ -2 \\ 3 \end{bmatrix}$.

Then, find a unit vector $\mathbf{u}$ in the direction of $\mathbf{v}$.

Distances in $\mathbb{R}^n$

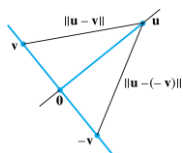
If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, the distance between $\mathbf{u}$ and $\mathbf{v}$ is the length of the vector $\mathbf{u} - \mathbf{v}$. Thus,

$$\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

Example 3

Two lines through the origin in $\mathbb{R}^2$ (or $\mathbb{R}^3$) determined by vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if and only if the distance from $\mathbf{u}$ to $\mathbf{v}$ is the same as the distance from $\mathbf{u}$ to $-\mathbf{v}$, as shown in the figure.

Use this to show that $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if and only if $\mathbf{u} \cdot \mathbf{v} = 0$.



Orthogonality

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are orthogonal if and only if

$$\mathbf{u} \cdot \mathbf{v} = 0$$

Equivalently, two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are orthogonal if and only if

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

Example 4

Calculate the distance between

$$\mathbf{u} = \begin{bmatrix} 12 \\ 3 \\ -5 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}.$$

Are $\mathbf{u}$ and $\mathbf{v}$ orthogonal?

Example 5

True or False.

The zero vector in $\mathbb{R}^n$ is orthogonal to every vector in $\mathbb{R}^n$.

Orthogonal Complements

If a vector $\mathbf{v}$ is orthogonal to every vector in a subspace W of $\mathbb{R}^n$, then $\mathbf{v}$ is orthogonal to W

The set of all vectors $\mathbf{v}$ which are orthogonal to W forms the orthogonal complement of W , denoted $W^\perp$ (and read as “ W perp”).

Orthogonal Complements

A vector $\mathbf{v}$ is in $W^\perp$ if and only if $\mathbf{v}$ is orthogonal to every vector in a basis of W .

Note that $W^\perp$ is a subspace of $\mathbb{R}^n$.

Example 6

Let W be a plane through the origin in $\mathbb{R}^3$, and let L be the line through the origin perpendicular to W . Describe $W^\perp$ and $L^\perp$.

Theorem

Let A be an $m \times n$ matrix. The orthogonal complement of the row space of A is the null space of A , and the orthogonal complement of the column space of A is the null space of A^T .
